
Updating Beliefs with Information

Conditional Probability, Independence & the Law of Total Probability

Prepared by Prof. Saurabh

New information changes the odds. A blood test comes back negative; a die is known to be even; we learn which forest we are standing in. This companion shows how a single, intuitive idea — *shrinking the world down to what we now know* — defines conditional probability, why some events ignore each other (independence), and how to assemble a tricky overall probability from clean, manageable pieces using the *law of total probability* and its trusty tool, the tree diagram.

What's inside

1	Probability changes when you learn something	2
2	Conditional probability: shrinking the world	2
3	The multiplication rule: chaining events together	4
4	Independence: when a clue tells you nothing	4
5	Total probability: assemble the whole from clean pieces	6
6	Practice problems	8

1 Probability changes when you learn something

In Session 2 every probability was assigned “cold,” before anything was known. But real life feeds us clues: a symptom appears, a test result arrives, we discover where we are. Each clue should *update* our probabilities. That updated number is a **conditional probability**.

What, why, where — for this whole session

What: how a probability *changes* once you learn that another event happened — plus a way to build a hard overall probability out of easy pieces. **Why:** almost every real decision is made *with* partial information, never in the dark; conditioning is how that information enters the maths. **Where:** medical testing, spam filters, reliability, quality control — and, next session, Bayes’ theorem, which is just conditional probability run in reverse.

2 Conditional probability: shrinking the world

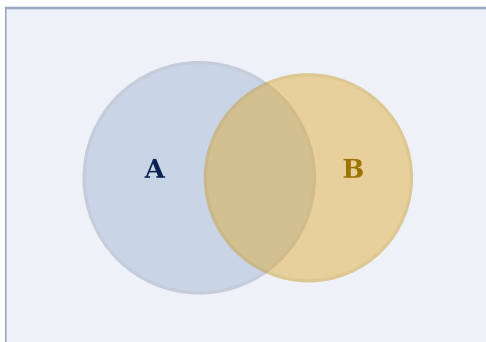
The idea. Asking “what is $P(B)$?” surveys the whole world S . Asking “what is $P(B \text{ given } A)$?” throws away every outcome where A did not happen, then asks: of the world that is *left* (just A), how much is also B ?

Zooming in

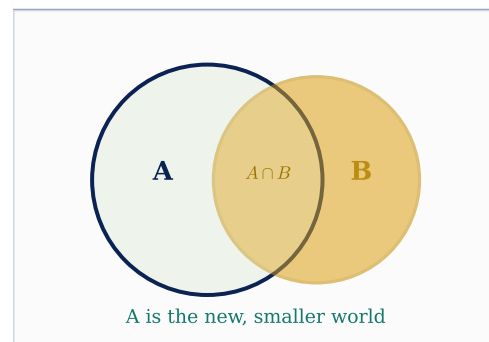
Conditioning is like zooming a map. Before, you saw the whole country (S). Someone tells you “we are in this state” (A). You discard the rest of the map and re-scale *this state* to fill the screen. Probabilities are re-measured against the new, smaller world — which is why the denominator becomes $P(A)$ instead of 1.

Conditioning on A = zooming into A and asking how much of it is also B : $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Before: full sample space S



After we learn A happened



Conditioning on A means deleting everything outside A and treating A as the new sample space. The conditional probability is the share of that shrunk world that also lies in B — the overlap $A \cap B$ divided by all of A .

Definition 2.1: Conditional probability

The probability of B given that A has occurred (read “ B given A ,” not “ B by A ”) is

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

The denominator $P(A)$ is the *size of the new world*; the numerator $P(A \cap B)$ is how much of B survives inside it.

Feel the formula

If A is half the world and the A -and- B overlap is one-tenth of the world, then *within A* the chance of B is $\frac{0.1}{0.5} = 0.2$. We didn’t change reality — we just changed the *reference frame* to “ A has happened,” and re-weighted B against it.

Conditioning = pick the right denominator (one column)

	Young	Middle-aged	Senior	Total
No default	10,503	27,368	259	38,130
Yes default	3,586	4,851	120	8,557
Total	14,089	32,219	379	46,687

$$P(\text{no default} | \text{middle-aged}) = \frac{27,368}{32,219} \approx 0.85 \text{ — just restrict to the gold column}$$

Conditioning, seen in a table. “Given middle-aged” tells us to ignore every other column and live inside the gold one; the conditional probability is then just one cell divided by that column’s total. Different “given” = different column = different denominator.

Worked example 2.1: A contingency table, read both ways (loan defaults)

A bank records age group versus loan default for 46,687 customers. Among the 32,219 middle-aged customers, 27,368 did *not* default; overall 38,130 customers did not default.

$$P(\text{no default} \mid \text{middle-aged}) = \frac{27,368}{32,219} \approx 0.85,$$

$$P(\text{middle-aged} \mid \text{no default}) = \frac{27,368}{38,130} \approx 0.72.$$

Same overlap on top, but *different denominators* — because the two questions zoom into different worlds (“the middle-aged” vs. “the non-defaulters”). Conditioning is, at heart, just choosing the right denominator. Beware: $P(B \mid A)$ and $P(A \mid B)$ are generally *not* equal.

3 The multiplication rule: chaining events together

Rearranging the conditional formula (clear the fraction) gives a rule for the probability that two — or many — things *all* happen.

Key formula 3.1: Multiplication rule

$$P(A \cap B) = P(A) P(B \mid A) = P(B) P(A \mid B),$$

and for a sequence it simply chains: $P(A \cap B \cap C) = P(A) P(B \mid A) P(C \mid A \cap B)$.

Walking a path, gate by gate

To pass through three gates in a row, multiply the chance of clearing each gate *given* you reached it. “And” across a sequence means *multiply along the path*; each step’s probability is conditioned on having survived the steps before it.

Worked example 3.1: Two letters from two bags

Bag 1 holds 5 cards, Bag 2 holds 6 cards. The picks are independent (different bags), so the chain just multiplies the single-pick chances. The probability of drawing a particular letter from each, for instance, is $\frac{1}{5} \cdot \frac{1}{6} = \frac{1}{30}$, and the probability that *both* cards are vowels is $\frac{3}{5} \cdot \frac{2}{6} = \frac{1}{5}$ (using the count of vowels in each bag).

4 Independence: when a clue tells you nothing

Sometimes learning A leaves the chance of B exactly as it was. Then the events are **independent**, and conditioning changes nothing.

Definition 4.1: Independent events

A and B are **independent** if knowing one tells you nothing about the other:

$$P(B | A) = P(B) \iff P(A \cap B) = P(A) P(B).$$

The right-hand form is the easy *test*: multiply the two single probabilities and see if you get the overlap.

Coins have no memory

A fair coin that just landed heads is no more (and no less) likely to land heads again — it has no memory. Successive flips are independent. The gambler who “feels a tails is due” is committing the classic fallacy of imagining memory where there is none.

With vs. without replacement — the practical switch

With replacement (you put the card back, the deck resets) keeps draws *independent*. **Without replacement** (the deck shrinks and “remembers” what you took) makes them *dependent*. Spotting which one a problem uses is half the battle.

Worked example 4.1: The same jar, two ways

A jar holds 5 red, 6 blue, 2 white marbles (13 total). Draw two; what is $P(\text{both red})$?

With replacement: $\frac{5}{13} \cdot \frac{5}{13} = \frac{25}{169} \approx 0.148$, **Without replacement:** $\frac{5}{13} \cdot \frac{4}{12} = \frac{5}{39} \approx 0.128$.

The clue “a red is already gone” lowers the second probability — dependence in action.

Worked example 4.2: Testing independence (airports)

Of 100 surveyed flights, 20 departed late. Airport A handled 45 flights, 9 of them late. Are “departed from A” and “departed late” independent? Compare:

$$P(\text{late}) = \frac{20}{100} = 0.20, \quad P(\text{late} | A) = \frac{9}{45} = 0.20.$$

They are *equal*, so the events **are independent** — which airport you left from told you nothing about lateness. (If the two numbers had differed, the events would be dependent.)

Worked example 4.3: Independent spam features

A model uses two *independent* features: the word “offer” ($P(A) = 0.6$) and a suspicious link ($P(B) = 0.4$). Then

$$P(\text{both}) = 0.6 \times 0.4 = 0.24,$$

$$P(\text{either}) = 0.6 + 0.4 - 0.24 = 0.76,$$

$$P(\text{neither}) = 1 - 0.76 = 0.24.$$

Independence let us multiply for “both”; the addition rule (with the overlap removed) gave “either”; and the complement trick finished “neither.”

5 Total probability: assemble the whole from clean pieces

Some probabilities are hard head-on but easy once we split the world into cases. If the cases $A_1, \dots, A_k$ are **mutually exclusive** and together cover everything (a *partition* of S), then any event B can be reached only through one of them.

Key formula 5.1: Law of total probability

If $A_1, \dots, A_k$ partition the sample space, then

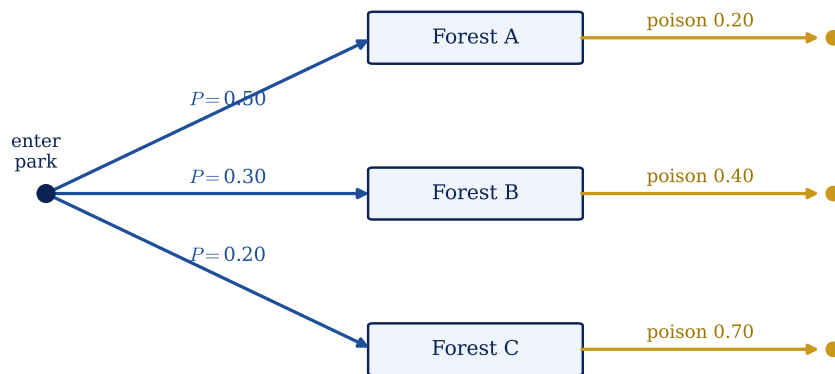
$$P(B) = \sum_{i=1}^k P(A_i) P(B | A_i).$$

In words: walk down every branch, multiply the probabilities along each path, then add the paths up.

Many roads into one town

To count travellers reaching a town, don’t track each person. Take every road leading in, count the traffic on each road, and add. Each “road” is a case A_i ; the traffic on it is $P(A_i) P(B | A_i)$. The total probability is the total traffic.

A tree splits a messy question into clean branches



$$\text{Total probability: } P(\text{poison}) = \sum_i P(\text{Forest}_i) P(\text{poison} | \text{Forest}_i) = 0.36$$

A tree makes total probability visual. Each first branch is a case (which forest?), each second branch is the event given that case (poisonous?). Multiply along a branch, then sum across branches.

Worked example 5.1: The poisonous-plant park

A park is 50% Forest A (20% poisonous), 30% Forest B (40% poisonous), 20% Forest C (70% poisonous). Pick a random plant. Each forest is one branch:

$$P(\text{poison}) = (0.20)(0.50) + (0.40)(0.30) + (0.70)(0.20) = 0.10 + 0.12 + 0.14 = 0.36.$$

No single forest is “36%”, yet the weighted blend across all three is.

Worked example 5.2: A noisy binary channel

A channel sends a 1 with probability 0.4 (and a 0 with 0.6). A sent 0 is received correctly 90% of the time; a sent 1, 95%. What is $P(\text{a 1 is received})$? A 1 can be received two ways — correctly from a sent 1, or wrongly from a sent 0:

$$P(\text{rec 1}) = \underbrace{(0.4)(0.95)}_{\text{sent 1, correct}} + \underbrace{(0.6)(0.10)}_{\text{sent 0, flipped}} = 0.38 + 0.06 = 0.44.$$

Worked example 5.3: Will the mining job finish on time?

Rain finishes the job on time with probability 0.42; no rain, 0.90. Rain has probability 0.45. Split on the weather:

$$P(\text{on time}) = (0.45)(0.42) + (0.55)(0.90) = 0.189 + 0.495 = 0.684.$$

6 Practice problems

Your turn

1. **(complement + independence).** Of 300 adults, 272 are right-handed. Picking 3 *with replacement*, find $P(\text{all three right-handed})$ and $P(\text{at least one right-handed})$. (Hint: “at least one” is easiest as $1 - P(\text{none})$.)
2. **(without replacement).** From a 52-card deck dealt without replacement, find $P(\text{both of the first two cards are queens})$. Redo it *with* replacement and compare — which is larger, and why?
3. **(independence test).** A card is drawn from a deck missing the ace of diamonds (51 cards). Are “clubs” and “queen” independent? (Test whether $P(\text{club})P(\text{queen})$ equals $P(\text{club and queen})$.)
4. **(total probability).** A user clicks an ad with probability 0.04 on desktop, 0.06 on tablet, 0.10 on mobile; device shares are 0.5, 0.2, 0.3. Find the overall click probability.

The thread to the next session

We now go *forward*: from a cause (A_i) to the chance of an effect (B). But the truly powerful question runs *backward*: we observe the effect (a positive test, the word “offer”) and want the probability of the hidden cause (disease, spam). Flipping that conditional is exactly what **Bayes’ theorem** does — the engine of Session 4.